



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.					
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Institute: (0203)SYMBIOSIS CENTRE FOR MANAGEMENT AND HUMAN RESOURCE DEVELOPMENT, PUNE
Programme: (020344) MASTER OF BUSINESS ADMINISTRATION (INFRASTRUCTURE DEVELOPMENT AND MANAGEMENT)
Batch: 2024-26
Semester: I
Course: Roads, Highways and Bridges
Course Code: 0203440114
Date: 15/03/2025
Day: Saturday
Maximum Marks: 60
Time: 09:30 - 12:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- 1) Use of scientific calculators is permitted.
- 2) All questions are compulsory.

- Q.1 What are the various categories of road maintenance, and how do they differ in terms of frequency and extent of repairs? (CO 5) (5 Marks)
- Q.2 Summarize the process of obtaining governmental approvals for highway projects. (CO 1) (5 Marks)
- Q.3 (a) Propose a toll fee structure for a 7-axle truck on a 250 crore bridge considering the following information: (CO 2) (5 Marks)

Cost of permanent bridge, bypass or tunnel (rupees in crore)	Car, Jeep, Van or Light Motor Vehicle	Light Commercial Vehicle, Light Goods Vehicle or Mini Bus	Truck or Bus	HCM, EME or MAV	Oversized Vehicle
10 to 15	5	7.5	15	22	30
For every additional rupees five crore or part thereof, exceeding rupees fifteen crore and up to rupees one hundred crore.	1	1.5	3	4.5	6
For every additional rupees five crore or part thereof, exceeding rupees one hundred crore and up to rupees two hundred crore.	0.75	1.15	2.25	3.4	4.5
For every additional rupees five crore or part thereof, exceeding rupees two hundred crore.	0.5	0.75	1.5	2.25	3

- (b) Appraise the role of the government versus private contractors in toll collection. Which system provides better service and efficiency?

(CO 2) (5 Marks)

Q.4

- (a) Explain the role of road safety audits in reducing accidents. How does it differ from crash investigations?

(CO 5) (5 Marks)

- (b) Describe the steps involved in conducting a road safety audit for a newly completed road project.

(CO 3) (5 Marks)

- Q.5 (a) How can the concept of shadow pricing be applied to quantify the environmental impact of a highway project? In the context of highway economic analysis, explain how the time value of money affects the evaluation of long-term projects.

(CO 5) (5 Marks)

- (b) The cost of improving an existing road, 25 km long, is Rs 4.00 lakh per km. Road user costs (with and without the improvements), accident costs (with and without the improvements) and maintenance costs (with and without the improvements) are given in below table (in lakhs) for a 10-year period after the completion of the improvements. Assuming a discount rate of 10 percent, find out whether the project is economically justifiable?

Year (t)	Road User Costs With Impr.	Road User Costs Without Impr.	Accident Costs With Impr.	Accident Costs Without Impr.	Maintenance Costs With Impr.	Maintenance Costs Without Impr.
0	-	-	-	-	-	-
1	105.5	126.5	1.1	3.1	3.5	2.5
2	110.3	132.2	1.1	3.1	3.5	2.5
3	115.8	138.9	1.2	3.5	3.5	2.5
4	121.6	145.8	1.2	3.7	3.5	2.5
5	127.6	153	1.3	3.8	3.5	2.5
6	134	161	1.3	4	3.5	2.5
7	140.7	168.9	1.4	4.2	3.5	2.5
8	147.8	177	1.5	4.4	3.5	2.5
9	155.1	186.2	1.6	4.7	3.5	2.5
10	162.9	195.2	1.6	4.9	3.5	2.5

(CO 5) (10 Marks)

- Q.6 (a) What are latest Rules and Regulations of NHAI (CO 4) (5 Marks)
- (b) What are advantages of PPP Model over conventional infrastructure development by Government (CO 4) (5 Marks)
- (c) What is objective of NHAI (CO 4) (5 Marks)
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